IT3011 - Theory & Practices in Statistical Modelling

Analysis Of Variance (ANOVA)

H.M. Samadhi Chathuranga Rathnayake

Analysis of Variance (ANOVA)

An extension of the pooled variance t-test for the difference between two means. Though the term mentions the word variance, we are testing differences in means. ANOVA separates total variation into variation within (random variation) and among the groups. So long as the ratio between within and among variation is large enough, we conclude that there are significant differences between groups.

Assumptions:

- Populations are normally distributed.
- Populations have equal variances.
- Samples are randomly and independently drawn.

One Way ANOVA

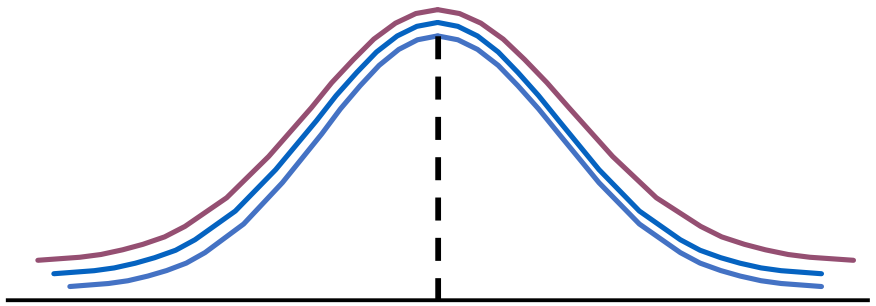
$$H_0\colon \mu_1 = \mu_2 = \cdots = \mu_c$$

All population means are equal. (no variation in means among groups)

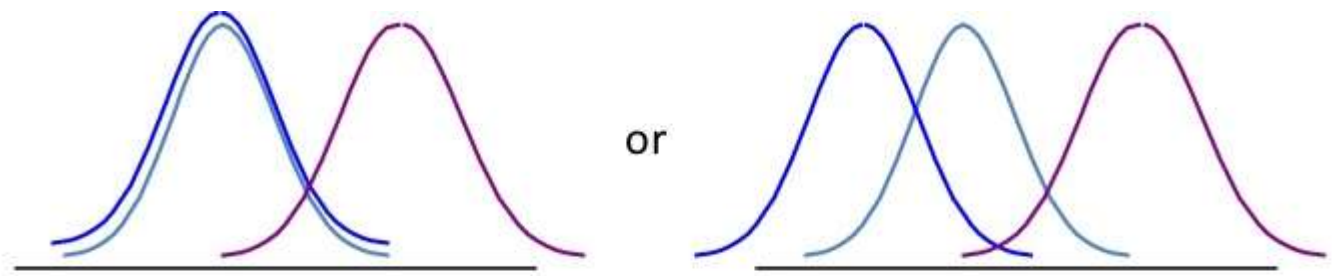
$$H_1\colon At\ least\ one\ population\ mean\ is\ different$$

At least one population mean is different. (not necessarily that all population means are different)

H_0



H_1



$\mu_1 = \mu_2 \neq \mu_3$ $\qquad$ $\mu_1 \neq \mu_2 \neq \mu_3$

One Way ANOVA

SST

Total sum of squares –

total variation

SST = SSA + SSW

SSA

Sum of squares among groups –

between group variation or variation due to factor

SSW

Sum of squares within groups –

within group variation or variation due to random error

One Way ANOVA

$$SST = \sum_{j=1}^{c} \sum_{i=1}^{n_j} \left(X_{ij} - \bar{\bar{X}}\right)^2$$

- c = number of groups or levels
- n_j = number of observations in group j
- X_{ij} = i^{th} observation from group j
- $\bar{\bar{X}}$ = grand mean (mean of all data values)

One Way ANOVA

$$SSA = \sum_{j=1}^{c} n_j \left(\bar{X}_j - \bar{\bar{X}}\right)^2$$

- $\bar{X}_j$ = sample mean from group j

$$Mean\ Square\ Among\ (MSA) = \frac{SSA}{c-1}$$

- where c-1 is the degrees of freedom of the SSA

One Way ANOVA

$$SSW = \sum_{j=1}^{c} \sum_{i=i}^{n_j} \left(X_{ij} - \bar{X}_j\right)^2$$

- $\bar{X}_j$ = sample mean from group j

$$Mean\ Square\ Within\ (MSW) = \frac{SSW}{n - c}$$

- Where n-c is the degrees of freedom of the SSW

One Way ANOVA Table

Source of variation	Degrees of freedom	Mean square	F
Between groups	c-1	$MSA = \frac{SSA}{c-1}$	$F_{stat} = \frac{MSA}{MSW}$
Within groups	n-c	$MSW = \frac{SSW}{n-c}$	
Total	n-c+(c-1)=n-1	$MST = \frac{SST}{n-1}$	

One Way ANOVA Test

The test statistic is the ratio of the mean of among variance and mean of within variance. The ratio must always be positive.

- c-1 is typically small
- n-c is typically large

Decision rule: Reject H_0 if $F_{stat} > F_{\alpha,c-1,n-c}$. Otherwise, do not reject.

One Way ANOVA Test

Ex:-

You want to see if three different tennis rackets hit the ball different distances. You randomly select five measurements from trials on an automated ball machine. At the 5% significance level, is there a difference in mean distance?

Racket 1	Racket 2	Racket 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

One Way ANOVA Test

Ex:-

When the needful calculations are done,

$$\bar{\bar{X}} = 227 \quad \bar{X}_1 = 249.2 \quad \bar{X}_2 = 226 \quad \bar{X}_3 = 205.8 \quad n_1 = n_2 = n_3 = 5 \quad n = 15 \quad c = 3$$

$$SSA = 4{,}716.4$$
$$SSW = 1{,}119.6$$

$$MSA = 4716.4/(3-1) = 2{,}358.2$$
$$MSW = 1119.6/(15-3) = 93.3$$

$$F_{stat} = \frac{2358.2}{93.3} = 25.275, \qquad F_{0.05,2,12} = 3.89$$

Since $F_{stat} > F_{0.05,2,12}$, Reject H_0 at 5% level of significance.

Conclusion: There is an evidence to say that at least one mean is different from the rest at 5% level of significance.

Checking Homogeneity of Variance

Levene's test is used for testing the assumption that the variances of each population are equal (assumption of ANOVA).

The test involves three steps:

1. Define the null and alternative hypothesis.

$$H_0\colon\ \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_c^2 \quad \& \quad H_1\colon Not\ all\ \sigma_j^2\ are\ equal\ (At\ least\ one\ variamce\ is\ different)$$

2. Compute the absolute value of the difference between each value and the median of each group.

3. Perform one-way ANOVA on these absolute differences.

Checking Homogeneity of Variance

Ex:-

Consider the above discussed example.

$$H_0\colon\ \sigma_1^2 = \sigma_2^2 = \sigma_3^2 \quad \& \quad H_1\colon Not\ all\ \sigma_j^2\ are\ equal$$

Racket 1	Racket 2	Racket 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

Colored cells are the median values.

Absolute difference from median		
Racket 1	Racket 2	Racket 3
3	7	4
12	9	18
10	8	7
14	0	2
0	11	0

Checking Homogeneity of Variance

Ex:-

$$F_{stat} = 0.92 \quad , \qquad F_{0.05,2,12} = 3.89$$

Since $F_{stat} < F_{0.05,2,12}$, Do not reject H_0 at 5% level of significance.

Conclusion: There is an evidence to say the homogeneity of variance assumption is satisfied at 5% level of significance.

Post – Hoc Analysis : Mean Comparisons

When the null hypothesis of ANOVA is rejected, it only suggests that there are significant differences between means. However, it does not specify which of the treatment means differ. Therefore further comparisons between group or treatment is required.

Post – Hoc Analysis : The Fisher Least Significant Difference (Fisher's LSD)

This is the most commonly used tool for testing the significance between all possible pairs of means. This method ensures error rate (Significance Level α) for each individual pair wise comparison.

The method is to declare the difference between any pair of means significant at $\alpha\%$ level if the,

$$|Observed\ Difference| > LSD$$

Where,

$$LSD = t_{df,\frac{\alpha}{2}}\sqrt{MSW\left(\frac{1}{n_j}+\frac{1}{n_i}\right)}$$

Where df=Error Degrees of Freedom

Post – Hoc Analysis : The Fisher Least Significant Difference (Fisher's LSD)

Ex:-

Doughnuts are supposed to absorb fat in various amounts, during cooking. In an experiment to compare 4 types of fat, 6 batches of Doughnuts were prepared from each type. The measurement was the amount (g) of fat absorbed. The data were analyzed as for a one-way treatment structure in a CRD. The following is the summary of data.

Treatment	Fat 01	Fat 02	Fat 03	Fat 04
r	6	6	6	6
mean	72	85	76	62

Post – Hoc Analysis : The Fisher Least Significant Difference (Fisher's LSD)

Ex:-

$$MSW = 100.9 \quad df = 20$$

So,

$$LSD = t_{20,\frac{0.05}{2}}\sqrt{100.9\left(\frac{1}{6}+\frac{1}{6}\right)} = 2.086 \times 5.80 = 12.1$$

Any pair with mean difference greater than 12.1 will be declared significant at 5% level of significance.

$$\bar{X}_{Fat1} = 72$$
$$\bar{X}_{Fat2} = 85$$
$$\bar{X}_{Fat3} = 76$$
$$\bar{X}_{Fat4} = 62$$

Ex:

$$|\bar{X}_2 - \bar{X}_1| = 85 - 72 = 13 > LSD$$

So the difference of the means of Fat 01 & Fat 02 is significant at 5% level of significance. We can check this for other pairs too.